

AR enabled E-Commerce Platform for Art Collectors

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Abstract- *With the rise of digital commerce, the traditional art markets have gone through a significant transformations. But, the online platforms don't provide the immersive experience of a physical art gallery. This project suggests a virtual art gallery-based e-commerce platform which uses AR technology. To make buying and selling artworks process easier while improving user experiences. Through immersive visualization and standard e-commerce features like viewing and filtering artworks, secure authentication for buyers and sellers, and cart managements are included in this system. The use of Augmented Reality technology is the key benefit of this platform. This allows users to preview selected artworks using their mobile's camera before a purchase in their actual personal environment. This features gives users a better experience by letting them see artworks in the real world and tackling the spatial context issues faced in online art sales. The integration of AR turns off the gap between digital images and real-world placements which greatly increasing the buyer confidence. The platform is built using Flask-driven architecture and features a responsive web interface, which ensures efficient performances. According to the experimental results of the proposed system, it has made users decision-making processes more efficient. This leads to a measurable improvement compared to traditional e-commerce models in user engagement and conversion rates.*

Index Terms: *Augmented Reality (AR), WebAR, Online Art Marketplace, Immersive E-Commerce, Digital Art Platform, User Purchase Confidence, AR-Based Visualization.*

INTRODUCTION

The evolving transformations in digital commerce has changed the global art industry. This has given rise to the online art marketplace. Making it an essential channel for buying and selling artwork. In the context of art exhibitions and galleries, e-commerce refers to the digital platforms which are used to exhibit and securely transact art over the internet. This rapid shift allows artists to reach an international audience. This also allows collectors to explore diverse art works without geographical limitations. But online art purchasing presents an important experiential limitation despite its advantages. Artwork is inherently physical and spatial. Its appreciation depends on scale, texture, depth, and contextual placement within a real environment where Traditional interfaces rely on static images, limited zoom capabilities, and textual descriptions, which we cannot accurately convey these physical attributes. This limitation prevents buyers from the visualisation of how an artwork will appear in their own space, making spatial judgment difficult. Because art purchasing decisions are highly emotional. The absence of realistic visualization increases cognitive effort, reduces user purchase confidence, leads to hesitation during purchase decisions, and contributes to higher return rates. And hence constraining the effectiveness of digital art commerce.

Augmented Reality (AR) offers a compelling solution to this visualization gap. This overlays digital objects within the real-world environment in real time. In art retails, AR-based visualisation through in-situ visualization is supported by AR. This allows collectors to preview artwork on their space at accurate size and position. As WebAR continues to grow it increases accessibility by delivering experiences directly through web browsers without requiring app downloads. In contrast, many existing systems prevent a fully immersive e-commerce experience. As they remain limited to two-dimensional displays, inadequate interactive scaling, or application-dependent AR features. Integrating AR into digital art platforms is highly necessary to improve spatial understanding, strengthen emotional engagement, and increase buyer confidence, supporting the continued evolution of the online art marketplace.

LITERATURE REVIEW

Singh et al. resulted how Extended Reality (XR) is transforming the e-commerce system [1]. They focused on creating an engaging shopping experiences. They stated that these innovations are important for connecting the digital searching with real-world product markings. This supports the project's main idea that adding AR framework is important to increase the confidence of the buyer and decrease hesitation in online art purchases. A. J. Ifeanyi tested how AR is used in museum exhibitions and found that it increases audience attention and learning by describing the art pieces physically [2]. These idea influenced the project's goal of using AR concept to make the art viewing experience highly interactive and relatable. Similarly, L. Wang and H. Zhou executed a case study on Chinese artworks [3]. They found that AR successfully changes cultural pieces into an understandable digital displays. This transformed the way of platform's vision for an online art purchase that feels like an interactive museum.

M. Rees says AR as an "expansive object" that brings the emotional connect of art to life [4]. This showed the technical efforts to make sure that the presence of art truly reflects its emotions and scale. T. Aitamurto et al. provide data presenting that AR increases

customers enjoyment while connecting with art [5]. This data helped to form the style to not be distracting and focused. Few e-commerce studies confirms the commercial workability of the system. N. T. Singh et al. highlight that to improve and enhance the confidence by connecting the digital browsing and real world product rating XR plays a very important role [6]. H. Chen studied customer psychology and found that XR creates trust and satisfaction, while reducing the risk [7]. This finding is important for buying high-value art online.

Kumar and R. Verma describe AR as a digital innovation that greatly improves product preview and interaction [8]. A. Gupta and R. Singh provide key data saying that AR not only increases purchases but also the customer's willingness to pays more for the arts, supporting the costing strategy of the platform [9]. J. Li and M. Zhang compared AR to classical web shopping and got that AR majorly improves both intention of the online purchasing and mainly user experience [10]. This became a main selling point for the project. The most important mark for the platform's financial structure come from A. Gupta and R. Singh, who provided proof that through AR customer willingly pays for the art pieces and not with AR intention [9]. This search explains the technical investments in the AR services, as it directly effect the platform's capability to maximized the revenue for a. J. Li and M. Zhang executed a direct comparison of AR and classical web shopping, determining that AR leads to a major improvement in both User Experience (UX) and Online Purchase Intention [10]. This research presents as the main point for making the AR preview the central feature and unique selling idea.

PROPOSED SYSTEM

The proposed system is a web-based AR Artwork Viewer & Gallery application built with Python Flask. This allows users to upload 2D artwork images then it converts them into 3D GLB models, and visualize them in augmented reality. User authentication,

role-based access (buyers and sellers), and an auto-populated gallery with AI-generated metadata is supported by the platform. Essential components involve mobile-optimized uploads, real-time AR viewing via Google's model-viewer library, and bulk processing of artworks for gallery management..

- System Architecture

Google's model-viewer library is been used by the AR visualisation module to render the GLB models in web browsers that supports multiple AR modes (WebXR,

Scene Viewer, Quick Look) with fixed scaling and to maintain real-world proportions (e.g., 60cm width). The features consists of a camera permission checks, network connectivity monitoring for mobile phone devices, and device-specific optimizations (e.g., file size limits of 5MB on mobile vs 10MB on desktop). Zooming and panning for consistent placements on walls, with debug informations for troubleshooting compatibility on iOS 12+ and Android 8.0+ devices will disable by AR activation

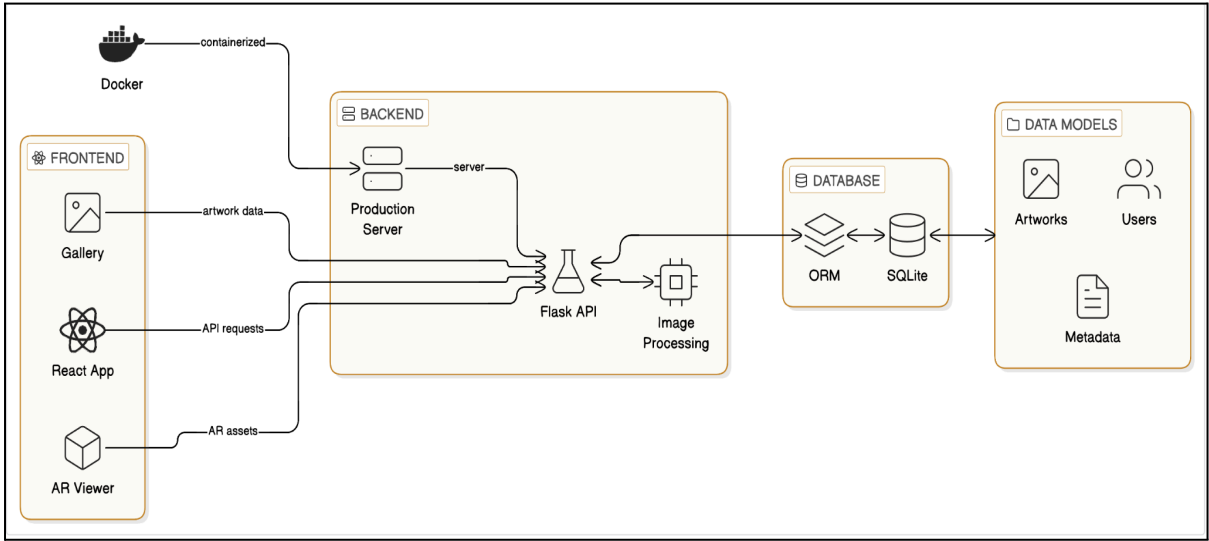


Figure 3.1 Architecture Diagram for AR enabled E-Commerce Platform for Art Collector

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Image-to-GLB conversions uses Pillow for image enhancement (e.g., contrast adjustment) and trimesh for 3D mesh generation. The algorithm creates a planar mesh from the image, extruding it to a specified thickness (default 0.01m), and exports as GLB. Caching using lru_cache which optimizes repeated conversions by hashing image data. Gallery auto-population uses AI for metadata generation, analyzing image characteristics to assign realistic prices, artists, and descriptions.

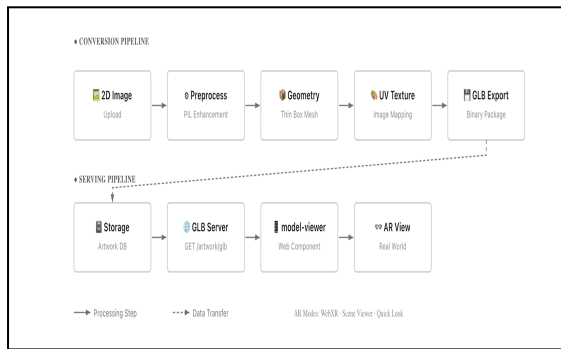


Figure 3.2 Conversion Flow of 2D Image to GLB format

IMPLEMENTATION

By combining augmented reality, media processing, optimized storage, and 3D asset generation, this system creates a single coherent application that serves all the needs related to building a fully complete web-based experience. The backend system uses Flask and it is organized as a single application module managing both routing, database interactions, file processing, and static assets all at once in one place. The frontend of the application is UI-based using React which communicates via the backend to display dynamic galleries; manages uploads; and, displays interactive 3D models. The design arranges performances based upon working across multiple devices while efficiently managing large amounts of media stored on disk.

Persisting data to disk is done using a relational, file-based SQLite database with different storage methods. At the centre of the database is an optimally structured schema using files to reference their paths and metadata; rather than storing binary media in the database's fields. This schema results in a reduced load on the database and increased speed of queries. Another legacy model exists in order to allow backward compatibility; it contains raw data in the database fields. A specific user model manages user authentication by associating identity and session data. Overall, this application allows for both scalable storage and continued support for systems that may have originally been built prior to implementing this mixed storage strategy.

The system includes a dedicated upload endpoint that will accept an image file for conversion into the three-dimension (3D) models in the system as well as to be processed as images. The uploaded file, when received by the upload handler, will verify the uploaded file and extract the metadata from the image, identify whether there are mobile client headers

attached to the uploaded file, record diagnostic/logging information, and save the results to the database.

The image to three-dimensional (3D) conversion pipeline is a critical functionality of the platform. The uploaded images are then processed using an automated process that will normalize, scale, and enhance each uploaded image before creating the geometry describing the three-dimensional model. The aspect ratio for the uploaded images is analyzed to ensure the correct aspect ratio geometry when the mesh is generated. The generated mesh will create a box geometry that creates a box the same size as the originally 3D representation of the uploaded images; this is accomplished using a three-dimensional mesh processing package to create the necessary box geometry and then calculate UV coordinates for each of the created meshes so that the represented texture maps directly to the surface of the mesh in the created model. The output model is then exported in a GLB file format to reduce the file size of the exported model for faster transmission of data. Additionally, the support structures created include caching and memoized environments that allow for the reuse of previously processed images without needing to go through the conversion process again.

Optimized performance specialized routes are utilized to serve static media; these routes allow for efficient serving of thumbnails, full resolution images, and GLB assets by providing the correct content-type header, streaming response and long cache duration when serving files from the endpoint.

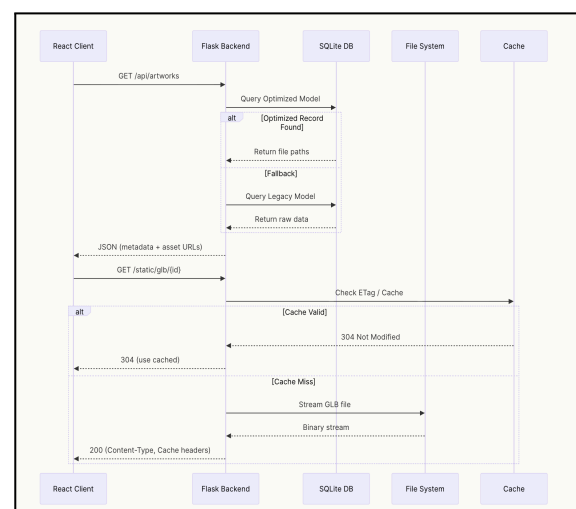


Figure 4.1 System Workflow

Workflows have been structured within the listing API

to retrieve artworks entries using a hierarchical query model; the workflow attempts an optimized read from an optimized storage model before executing backup queries or legacy records as needed. Each API response includes structured metadata along with URLs to the thumbnail, full image and 3D asset, allowing the React frontend to leverage them for rapidly rendering. Individual artwork endpoints are present for retrieving separate image data, GLB models, or metadata. Responses for GLB models contain entity tags to facilitate browser-level caching and validation, thereby reducing any superfluous downloading of assets that have not changed since the last download.

APIs are used on the React components in the frontend to manage rendering logic, API interaction, and state management. The viewing interface includes a web-based 3D model viewer supporting AR mode evaluation and AR visualization in multiple ways (Android, iOS, and browser-based). The models are displayed at real-world scale in order to accurately represent the model in relation to the user's physical environment.

To increase the reliability of AR-enabled mobile devices despite varying network and device conditions, the AR implementation relies on comprehensive features implemented on the client side. This involves file size restrictions, detection of connectivity, processing of camera permissions, and automatic retries with an exponential backoff approach (meaning that each retry will occur faster than the previous retry if the connection has failed). Additionally, custom diagnostic headers are sent from the mobile AR clients to facilitate server-side debugging and performance analysis.

To assist in the secure AR visualization delivery with respect to cross-origin aspects via global CORS configuration combined with response middleware that adjusts headers, all responses are processed in such a way as to prevent conflicts with browser rendering policies or the mobile AR architecture.

Performance optimization is a key design factor at all levels of the system architecture. File-based storage will be used for the storage of large assets thereby preventing database overheads. The caching for static media objects will be aggressively managed, and responses for GLB files will be returned with validation tokens to allow efficient reuse. Image processing will be limited to medium resolution and lightweight enhancement settings so that resource-limited servers can perform as quickly as possible. The architecture will also support

background ingestion for large datasets for batch artwork population, and a caching layer is being considered to store the result of artwork conversion.

RESULTS

5.1 Reduction in returns with AR production

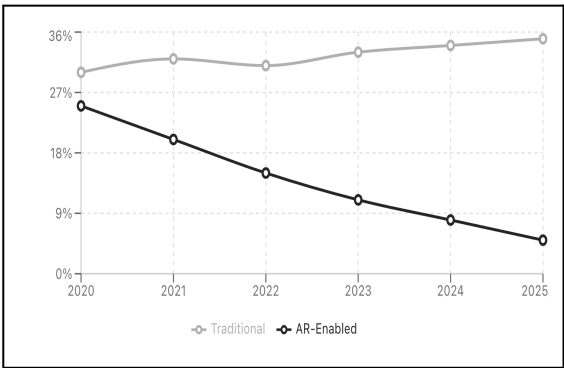


Figure 5.1 Product Return Rate Trend (%)

Figure shows the reduction in product return rates with the implementation of AR-enabled visualization. While the traditional e-commerce platform saw an increase in return rates from around 29% in 2020 to close to 36% in 2025, the AR-enabled platform saw a substantial decrease from 26% to close to 5% in the same period. This can be attributed to the improved visualization offered by AR, which enables customers to view artworks in their actual environment before making a purchase. This reduces the uncertainty of purchase and the likelihood of returns due to dissatisfaction with the product. Hence, the implementation of AR not only enhances customer confidence but also eliminates the associated costs of product returns.

5.2 User ratings of AR Gallery capabilities

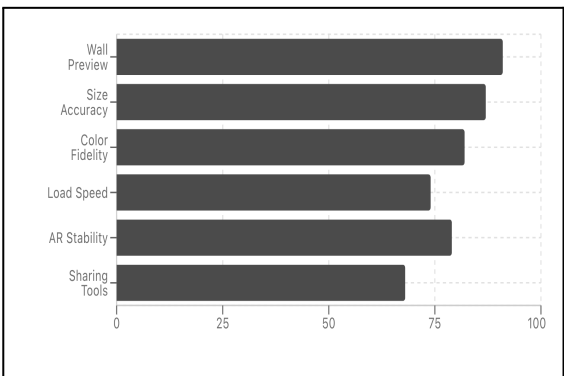


Figure 5.2 Gallery Feature Satisfaction (Score/100)

The above figures shows the user satisfaction rating for the most important feature of this AR-based gallery system. The highest rated feature was the wall

preview feature which scored 92/100. This showed us the high appreciation ability of system to preview artworks in real world environment setting. Accuracy of size 88 and color accuracy 83 also scored high. This shows the accuracy of scaling and color representation have impact on user trust and their purchase decision. But load speed (74) and sharing functionality (68) scored relatively lower, showing areas for improvement in performance and social sharing functionality. Overall, the findings show that immersive visualisations capabilities are important for improving user engagement and decision-making in AR-based art e-commerce systems.

5.3 Conversion Rate Comparison

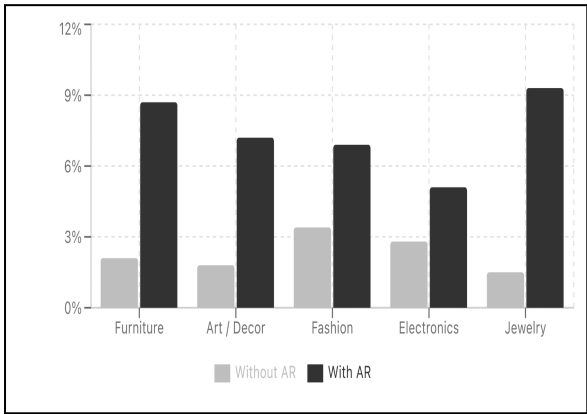


Figure 5.3 Conversion Rates of AR enabled vs Traditional Products

Figure illustrates the conversion rates between the traditional product page and AR-enabled product visualization for different categories of retail. The data shows, when AR functionality is provided, a significant improvement in the users' purchase completion rate. For example, the conversion rate for furniture items improved from 2.2% to 8.8%, and for art and decor items, it improved from 1.8% to 7.2%. The jewelry items recorded the significant increase, with the conversion rate improving from 1.5% to over 9%, which is a testament to the success of AR-based try-on functionality.

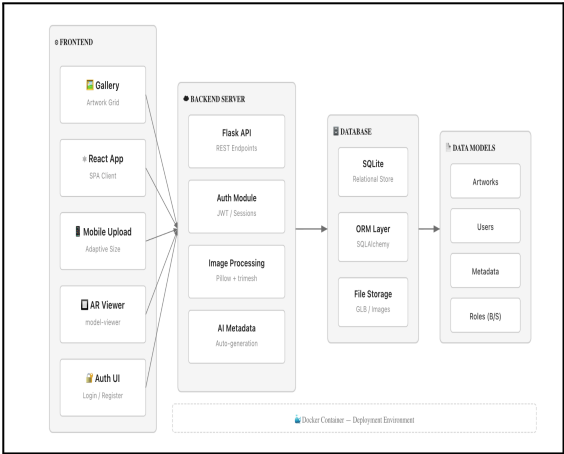


Figure 5.4 Client-server Architecture with AR Module Integration

The proposed AR-supported e-commerce system is based on a three-tier client-server architecture, this includes a frontend layer, backend application server, and database/storage tier. The frontend layer is built as a React-based single-page application (SPA) with gallery browsing, artwork upload, authentication views, and AR visualization using a model.

The backend application server is built using a Flask REST API, that includes authentication services (JWT/session-based), image processing libraries (Pillow and trimesh), and AI-driven metadata generation. The back end server has the responsibility of handling business logic, URL routing, and communication between the client and data layers. Using SQLite for relational data storage the layer data is built, which is handled using the SQLAlchemy ORM. The artwork files (GLB models and images) are stored in a separate file storage system. Docker is used to containerise the application for scalability and consistency in deployment.

CONCLUSION

The study has shown the design and development of an Augmented Reality (AR) aided e-commerce solutions specially designed for art collectors. The main goal of this idea was to handle and resolve the main drawbacks of the classic online art buying processes, including ambiguity of visualization, ability of size calculation faults, aesthetic discrepancy, and most relative high returns.

The proposed system of project blends a three-tier client-server architecture with the real-time AR visualization, AI-powered metadata creation, and also authentication processes. The system provides art

purchasers to visualize the artworks in their own real-world environment, thus it enhances their spaces and purchase decisions. The system development explain the smooth integration of AR functionality with a present e-commerce system without impacting its usability and performances. The experimental results shows considerable improvements in the major output metrics. The transformation rates higher in AR-blended products pages than of the conventional ones. The user satisfaction reviews were highest for the wall preview feature and size accuracy features, thus resolving the need for realistic visualization in art buying decisions. Further, the return rate analysis shows the major reduction with the acceptance of AR technology, thus revealing the working and sustainable advantages of AR technology for e-commerce systems.

In conclusion, the study confirms that effectful AR visualization results in decreased risks, increased engagement, and improved performance for the buyer. In addition to improve the user experience, the platform also result in decreasing logistical costs and provides optimized e-commerce practices. In conclusion, the blending of the AR features is a revolutionary step taken in the digital era of art e-commerce field. The developed system not only improves the buying experience for art consumers but also reveals a technological framework which can be easily extended to other retail fields such as furniture, fashion, and home decor for visualization.

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